EXECUTIVE SUMMARY

Improving Term Deposit Conversion: A Data-Driven Telemarketing Strategy for a Retail Bank

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Business Problem

A retail bank conducted 41,188 telemarketing contacts to promote term deposits but achieved a subscription rate of only 11.27%. Without identifying which customers are more likely to subscribe and which campaign practices are associated with better outcomes, the bank risks allocating its outreach capacity inefficiently and missing potential subscription opportunities.

Business Objective

Identify the customer segments, communication channels and campaign characteristics associated with higher subscription rates, and translate the findings into actionable recommendations for improving customer prioritisation and campaign resource allocation.

Methodology

Business Understanding → Data Audit → Data Preparation → Exploratory Analysis → Insight Generation → Business Recommendations → Dashboard Development

SQL Server (T-SQL) was used to prepare and analyse the data through six thematic areas:

- Campaign performance
- Customer segmentation
- Financial profile analysis
- Campaign strategy evaluation
- Macroeconomic indicators
- Multidimensional analysis

The findings were consolidated into an interactive Power BI dashboard to support campaign planning and decision-making.

Key Findings & Business Implications

1. Previous campaign success is the strongest prioritisation signal.

Customers with a previous successful campaign outcome achieved a 65.11% conversion rate, compared with 11.27% overall. Although they represented only 3.33% of contacts, they generated 19.27% of all subscriptions.

Business implication: Prioritising eligible customers with a history of successful responses could help the bank identify promising leads before applying demographic filters.

2. Customers aged 60+ represent a high-conversion segment.

Customers aged 60+ achieved a 39.56% conversion rate, while retirees achieved 25.23%. Both rates exceeded the overall campaign average.

Business implication: These groups warrant targeted outreach tests, although their smaller contact volumes mean they should complement rather than replace broader customer targeting.

3. Cellular is the better-performing contact channel.

Cellular contacts achieved a 14.74% conversion rate, compared with 5.23% for telephone contacts, and generated 83.04% of all subscriptions.

Business implication: Cellular deserves priority in campaign testing, although its cost-effectiveness must be verified before reallocating all outreach capacity.

4. Repeated contact attempts are associated with lower conversion.

Customers contacted once had a 13.04% conversion rate, compared with 5.49% among those contacted six or more times.

Business implication: The bank should test contact-frequency limits and assess whether additional outreach capacity would generate better results when directed towards new prospects.

5. Housing and personal loan ownership offer limited segmentation value.
Conversion rates differed only modestly according to housing and personal loan ownership.
Business implication: These variables should not be used as primary criteria for prioritising campaign contacts.
6. Interest-rate conditions may provide a campaign-planning signal.
Lower Euribor rates were associated with higher subscription rates in the observed data.
Business implication: The bank should monitor interest-rate conditions but gather further evidence before using them to determine campaign timing.

Business Recommendations

1. Prioritise previous campaign responders before applying demographic filters, subject to verifying their current eligibility.
2. Focus outreach on customers aged 60+ and retirees, while monitoring their conversion rates and subscription volumes.
3. Adopt a cellular-first contact strategy, subject to testing its performance and cost-effectiveness against telephone outreach.
4. Limit repeated contact attempts and redirect outreach capacity towards new prospects where testing demonstrates better results.
5. Remove housing and personal loan ownership from primary targeting rules, given their limited differences in observed conversion rates.
6. Monitor interest-rate conditions as an early campaign-planning signal while collecting additional evidence.

Business Impact

The analysis identifies opportunities to improve how the bank prioritises customers and allocates telemarketing capacity. In particular, previous campaign success provides a strong basis for identifying promising leads, while channel and contact-frequency findings offer practical areas for controlled testing.
Implementing and evaluating these recommendations could help the bank improve subscription conversion and make more informed campaign allocation decisions.

At a Glance

Metric	Result
Records analyzed	41,188 customer contacts
Variables analyzed	21
Baseline conversion rate	11.27%
Highest-converting segment	Previous responders — 65.11%
Strongest demographic segment	Age 60+ / Retirees 39.56%
Best-performing channel	Cellular 14.74% vs. 5.23% landline
Tools	SQL Server (T-SQL), Power BI

Full analysis, SQL scripts, Power BI dashboard, and documentation:
https://github.com/FARAHBENSALEM01/Bank-Marketing-Campaign-Analysis